

## Practice Problems on Qubit Operations and Quantum Logic Gates

1. Perform the following qubit operations:

a)  $\sigma_X|0\rangle, \sigma_X|1\rangle, \sigma_X|+\rangle, \sigma_X|-\rangle, \sigma_X|i\rangle, \sigma_X|-i\rangle$  where,  $\sigma_X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  and  $i = \sqrt{-1}$  as usual.

b)  $\sigma_Z|0\rangle, \sigma_Z|1\rangle, \sigma_Z|+\rangle, \sigma_Z|-\rangle, \sigma_Z|i\rangle, \sigma_Z|-i\rangle$  where,  $\sigma_Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Interpret the results obtained in both the sub-questions.

2. Perform the Hadamard gate operation on all the qubits as above and interpret the results. Given

$$H = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

3. Pauli Y-gate is given by  $\sigma_Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ . Show that  $\sigma_Y|q\rangle = i\sigma_X\sigma_Z|q\rangle$  for all six state of  $|q\rangle$  as in problem number 1.

4. The operation of CNOT gate is given by: “Flip the data bit when control bit is high (i.e.  $|1\rangle$ ). From this logic, generate the truth table for CNOT gate for the operation:  $\text{CNOT}|q_1q_2\rangle$ , where  $|q_1\rangle$  is control bit and  $|q_2\rangle$  is data bit. Consider following states:  $|q_1\rangle = 0, 1$  and  $|q_2\rangle = 0, 1, +, i$ .

5. Determine output of the following quantum circuit (X is Pauli-X gate) for all six inputs as in problem number 1:



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